

IVANA BARNES

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EDUCATION

University of Michigan – Ann Arbor
M.S. in Ecology and Evolutionary Biology

Ann Arbor, MI
August 2026

- GPA: 4.0
- Advisor: Dr. Gideon Bradburd
- Honors and Awards: Outstanding GSI Award (Departmental Teaching Award)
- Relevant coursework: Evolution, Population and Community Ecology, Research Ethics, Supervised Teaching, Programming in R, Statistical modeling in R.

Yale University
B.S. in Ecology and Evolutionary Biology

New Haven, CT
May 2023

- GPA: 3.98; *summa cum laude*
- Thesis Advisor: Dr. Thomas Near
- Honors and Awards: Edgar J. Boell Prize for excellence in biology
- Relevant Coursework: Evolutionary Biology, Phylogenetic Biology, Evolutionary Genetics, General Ecology, Ichthyology (+ Lab), Plant Diversity and Evolution (+ Lab), Evolutionary Architects, Data Analysis
- Other Activities
 - Assistant Musical Director of Whim ‘n’ Rhythm (Yale Senior acapella group)
 - Musical Director of Something Extra (Yale acapella group)
 - Member of Latina Women at Yale
 - Member of Yale Ecology and Evolutionary Biology Undergraduate Group (YEEBUG)

PUBLICATIONS

Massatti, R., Faske, T.M., **Barnes, I.M.**, Leger, E.A., Parchman, T.L., Richardson, B.A., and Knowles, L.K. (2025). Suturing fragmented landscapes: Mosaic hybrid zones in plants may facilitate ecosystem resiliency. *The Proceedings of the National Academy of Sciences*.
<https://doi.org/10.1073/pnas.2410941122>

Schmidt, S.A., Schmetterling, D.A., **Barnes, I.**, and Adams, S. B. (2023). The first occurrence of the southern plains crayfish (*Procambarus Simulans*) documented in Montana, USA. *Bioinvasions Records*.
https://www.reabic.net/journals/Bir/2023/1/BIR_2023_Schmidt_etal.pdf

Daiy, K., Frieson, T., Palacio-Londono, L., **Barnes, I.**, Olmedo, S., Aranda, P., Cirigliano, F., & Valeggia, C. (2022). Body norms, body image, and media in a market-integrating indigenous population in Argentina: A mixed-methods investigation. *American Journal of Human Biology*, e23813. <https://doi.org/10.1002/ajhb.23813>

RESEARCH EXPERIENCE

Gideon Bradburd Lab

Graduate Student

University of Michigan – Ann Arbor

January 2025 – present

- Research Project: Developing an update to BEDASSLE, an R package that uses MCMC in RStan to test for patterns of isolation by environment (IBE) in spatial genetic datasets while controlling for isolation by distance (IBD).

Lacey Knowles Lab

Graduate Student

University of Michigan – Ann Arbor

August 2023 – January 2025

- Research Project: Investigating the effects of area and least-cost distance between sky islands on patterns of genetic diversity in *Melanoplus oregonensis* grasshoppers.

Thomas J. Near Lab

Undergraduate Research Assistant

Yale University

May 2021 – May 2023

- Senior Thesis: Molecular Phylogenetic Analysis of Crayfish in the Southeastern US
- Inferred phylogenies for >300 crayfish specimens from mtDNA and ddRADseq data I collected
- Contributed to a manuscript announcing the presence of an invasive crayfish species in Montana
- Field work to collect *Cambarus rusticiformis* specimens

Reproductive Ecology Lab

Undergraduate Research Assistant

Yale University

May 2020 – August 2020

- Investigated the effects of social media on body image in the Toba people of Argentina
- Translated interviews from Spanish to English
- Analyzed qualitative interview data using Nvivo coding software
- Summarized key themes from the interviews to inform discussion in the manuscript

TEACHING EXPERIENCE

Graduate Student Instructor

University of Michigan – Ann Arbor

- Genetics Summer 2026
- Evolution (Graduate Level) Winter 2026
- General Ecology Fall 2025
- Statistical Model Building in R Winter 2025
- Introductory Biology: Ecology and Evolution Fall 2024, Winter 2024
- Introductory Biology Lab Fall 2023

PROFESSIONAL TRAINING

Workshop on Molecular Evolution

The Marine Biological Laboratory

Summer 2024

Woods Hole, MA

PRESENTATIONS AND POSTERS

Barnes, I.B. (2023). Molecular Phylogenetic Analysis of *Cambarus* Crayfish. [Presentation]. Yale EEB Senior Symposium. 2 May.

Barnes, I.B., Ghezelayagh, A., Simmons, J.W., Orr, O.D., Williams, C.E., & Near, T.J. (2022). Mitochondrial introgression in crayfish species *Cambarus rusticiformis* and *C. tenebrosus*. [Poster]. Yale Undergraduate Research Association Fall Symposium. 12 November.

Barnes, I.B., Ghezelayagh, A., Simmons, J.W., Orr, O.D., Williams, C.E., & Near, T.J. (2022). Mitochondrial introgression in crayfish species *Cambarus rusticiformis* and *C. tenebrosus*. [Poster]. Southeastern Fishes Council 48th Annual Meeting. 17 November.

FELLOWSHIPS AND GRANTS

Yale College Dean's Science Fellowship (4300 USD) Summer 2022
Funded 10 weeks of summer research, during which I performed fieldwork and collected molecular data for my Senior thesis.

Richter Summer Fellowship (1500 USD) Summer 2022
Funded an additional 3 weeks of research for the same summer.

Peabody Summer Internship (3000 USD) Summer 2021
Funded 10 weeks of research.

SKILLS

Analytical Skills

- Advanced R
- Basic Python
- Stan
- SLiM evolutionary simulation
- Tree sequence analysis in python with Tskit
- IQ-TREE
- Ipyrad
- Basic environmental niche modeling (ENM)
- QGIS

Lab Procedural Skills

- PstI/MspI ddRAD sequencing protocols
- DNA extraction and purification (SPRI beads, ethanol precipitation, etc.)
- DNA quantification (Qubit) and agarose gel electrophoresis

- PCR